

RESEARCH ARTICLE

MILP Modeling and Optimization of Multi-Objective Three-Stage Flexible Job Shop Scheduling Problem With Assembly and AGV Transportation

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ABSTRACT In the real manufacturing environment, the machining stage of the jobs and the assembly stage of the products are often completed in different workshops. In addition, automatic guided vehicle (AGV) plays an indispensable role in the transportation of jobs from machining workshop to assembly workshop. This paper studies multi-objective three-stage flexible job shop scheduling problem (FJSP-T-A) with minimizing both the makespan and the total energy consumption. In FJSP-T-A, jobs are first machined in flexible job shop, then are transported to assembly workshop by AGVs, and finally are assembled in assembly workshop. To solve this problem, a mixed-integer linear programming model (MILP) is developed and the optimal Pareto front for small-scale instances are solved by using the ϵ -method. FJSP-T-A is NP-hard, and an efficient multi-population co-evolutionary algorithm (MPCEA) is proposed to efficiently solve large-scale instances. In the MPCEA, we design a strategy to select relatively high-quality individuals to enhance the algorithm's convergence speed, and design a multi-objective variable-neighborhood search (MOVNS) method to improve the local search ability. Experiments are conducted to prove the effectiveness of the MILP model and the MPCEA.

INDEX TERMS Three-stage flexible job shop scheduling problem, mixed-integer linear programming, co-evolutionary algorithm, variable-neighborhood search.

I. INTRODUCTION

With the increasing concern for sustainable development in society, energy conservation and emission reduction have become one of the pressing issues in various industries. In the manufacturing industry, many equipment operations, production processes and material transportation consume large amounts of electricity and other forms of energy. The purchase of energy-saving equipment is a simple and

direct measure to reduce energy consumption, but it requires the investment of a large amount of money, which can be unaffordable for many small and medium-sized enterprises. Therefore, in the field of intelligent manufacturing, energy-saving scheduling has attracted the attention of more and more researchers with the advantages of short cycle time and small investment, and it has become a prominent research area within the scope of intelligent manufacturing [1], [2]. Flexible job shop scheduling problem (FJSP) widely encountered in real production. Under the background of energy conservation and emission reduction, studying

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energy-saving scheduling problems has become a valuable research topic [3], [4], [5], [6], [7], [8], [9].

In the real manufacturing environment, a product is usually composed of several parts. To complete a product, materials are first machined in machining workshop, then are transported to assembly workshop by AGVs, and finally are assembled in assembly workshop. Therefore, the study of the integration of machining, transportation and assembly is accordance with the real production and has been attracting increasing attention [10], [11], [12]. Compared with traditional shop scheduling, the integrated scheduling of machining, transportation and assembly involves more sub-problems and is much more complex.

In this paper, we study the three-stage flexible job shop scheduling problem (FJSP-T-A). In FJSP-T-A, jobs are first machined in flexible job shop (machining stage), then are transported to assembly workshop by AGVs (transporting stage), and finally are assembled in assembly workshop (assembly stage). Moreover, the capacity of the AGV is limited. The assembly workshop is a parallel machine shop. Specifically, FJSP-T-A should determine six sub-problems: operations sequencing and processing machine selection in machining stage, jobs sequencing and AGV selection in transporting stage, and products sequencing and assembly machine selection in assembly stage.

The objective is minimizing both the makespan and the total energy consumption. To solve this problem, a mixed-integer linear programming model (MILP) is developed and the optimal Pareto front for small-scale instances are solved by using the ε -method. Moreover, an efficient multi-population co-evolutionary algorithm (MPCEA) is proposed to efficiently solve large-scale instances. In the MPCEA, a strategy is designed to select relatively high-quality individuals to enhance the algorithm's convergence speed, and a multi-objective variable-neighborhood search (MOVNS) method is designed to improve the local search ability. The main contributions are as follows:

- (1) A mixed linear programming model is proposed for solving small-scale instances.
- (2) A multi-population co-evolutionary algorithm (MPCEA) is proposed for solving large-scale instances.
- (3) In MPCEA, a strategy of selecting relatively good individuals is designed to improve the convergence speed of the algorithm, and an efficient MOVNS is designed to improve the local search ability of the algorithm.

The rest of the paper is as follows: Section II presents the literature review. Section III develops the MILP model. Section IV describes the proposed MPCEA. Section V shows the results of the comparison experiment. Section VI provides the conclusion of the paper and an outlook for future research.

II. LITERATURE REVIEW

The existing literatures are reviewed from three aspects, namely multi-objective energy-saving FJSP, FJSP with AGV and FJSP with assembly.

A. LITERATURE REVIEW OF MULTI-OBJECTIVE ENERGY-SAVING FJSP

To minimize the total energy consumption, makespan and instability, Caldeira et al. [13] proposed an improved backtracking search algorithm. To minimize the makespan and total energy cost, and maximize the total availability, Mokhtari and Hasani [14] developed a multi-objective optimization model and proposed an improved evolutionary algorithm with incorporating a global criterion. Jiang and Deng [15] presented a kind of bi-population based discrete cat swarm optimization algorithm to minimize the sum of the energy consumption cost and the earliness/tardiness cost. Luan et al. [16] presented an enhanced non-dominated sorting genetic algorithm II to minimize the makespan, total delay time and total energy consumption. To minimize the makespan and total energy consumption, Lu et al. [17] proposed a discrete water wave optimization algorithm, Luo et al. [18] presented a knowledge-driven two-stage memetic algorithm, and Meng et al. [19] proposed an efficient multi-objective hybrid shuffled frog-leaping algorithm.

B. LITERATURE REVIEW OF FJSP WITH AGV

For the existing research about FJSP with considering transportation, AGV is used to transporting a job from machines to machines of different operations. Moreover, AGV can transport only one job at the same time. To minimize the makespan, Yao et al. [20] proposed a novel MILP model, Han et al. [21] presented a dual population collaborative genetic algorithm, Xin et al. [22] designed a multi-view modeling-based hybrid algorithm consisting of an estimation of distribution algorithm and an ant colony optimization, Luo et al. [23] proposed a cooperative hybrid EA, Zhang et al. [24] proposed a deep reinforcement learning method, Meng et al. [25] presented an improved genetic algorithm, and Pan et al. [26] presented a learning-based multi-population evolutionary optimization. To minimize the makespan, total AGV running time and total machine load, Wen et al. [27] proposed an effective hybrid algorithm. To minimize the makespan and total energy consumption, Zhang et al. [28] designed a deep Q-network-based memetic algorithm, Yao et al. [29] presented a multi-objective memetic algorithm with deep Q-network, and Dai et al. [30] proposed an enhanced genetic algorithm. With considering job cancellation, machine breakdown and AGV breakdown, Chen et al. [31] presented a Q-Learning-based NSGA-II algorithm.

C. LITERATURE REVIEW OF FJSP WITH ASSEMBLY

To minimize the makespan, Zhang and Wang [32] developed several dispatching rules with machine feedback mechanism, Fattahi et al. [33] proposed a new hybrid particle swarm optimization and parallel variable neighborhood search algorithm, Johnson et al. [34] designed a double DQN-based algorithm, and Daneshamooz et al. [35] presented two new algorithms based on variable-neighborhood

search. Zhang et al. [36] considered lot streaming and presented a distributed ant colony system to minimize the makespan, total tardiness and total workload. With the objective of minimizing the makespan and energy consumption, Ren et al. [37] proposed a novel heuristic algorithm integrated particle swarm optimization and genetic algorithm, and Hu et al. [38] proposed a learning-oriented multi-objective artificial bee colony algorithm and designed various operators to improve the performance of MLABC. With considering component sharing, Sun et al. [39] proposed a hybrid non-dominated sorting genetic algorithm with tabu search to minimize the makespan, total tardiness and count of machine changes.

In conclusion, several existing researches have studied the multi-objective energy-saving FJSP, FJSP with AGV and FJSP with assembly. The AGV is single-capacity and transports one job at a time. However, no research has focused on the energy-saving FJSP-T-A with simultaneously considering AGV transportation and assembly. Moreover, the AGV with multi-capacity is considered, and it can transport a certain number of jobs at the same time.

III. MILP MODEL

A. NOTATIONS

The notations definition is given in Table 1.

B. PROBLEM DESCRIPTION

As shown in the following figure, in FJSP-T-A, jobs are first machined in flexible job shop (machining stage), then are transported to assembly workshop by AGVs (transporting stage), and finally are assembled in assembly workshop (assembly stage). The assembly workshop is a parallel machine shop. Specifically, FJSP-T-A should determine six sub-problems: operations sequencing and processing machine selection in machining stage, jobs sequencing and AGV selection in transporting stage, and products sequencing and assembly machine selection in assembly stage. The objective of this paper is to minimize both the makespan and the total energy consumption. The entire production process is shown in Figure 1.

In processing stage, there are n workpieces are processed on m processing machines. Each operation can be processed on several processing machines. Each job contains several operations, which must be machined according to specific route. The processing machine has two modes, namely processing mode and idle mode. In processing mode, the machine processes a job. In idle mode, the machine waits for job after processing the current one.

In transportation stage, there are nA identical AGVs. For each job, it can be transported only when it is finished in processing stage. Each job can only be transported by one AGV. Moreover, the AGV is multi-capacity and can transport a certain number of jobs at the same time. The AGV has two modes, namely loaded mode and unloaded mode. In loaded mode, the AGV transport jobs from the processing shop to

TABLE 1. Notations definition.

Notations	Definition
i, i'	Job indices
n	Total number of jobs
I	Set of jobs and $I = \{1, 2, \dots, n\}$
p, p'	Product indices
np	Total number of products
P	Set of products and $P = \{1, 2, \dots, np\}$
PI_p	Set of jobs that product p contains
j, j'	The operation indices in the processing stage
n_i	Number of operations of job i in the processing stage
J_i	Set for the operations of job i and $J_i = \{1, 2, \dots, n_i\}$ in the processing stage
k, k'	Machine indices in the processing stage
m	Total number of machines in the processing stage
K	Set of machines and $K = \{1, \dots, m\}$ in the processing stage
$m_{i,j}$	Total number of machines in the processing stage that can process operation $O_{i,j}$
$K_{i,j}$	Set of machines in the processing stage that can process operation $O_{i,j}$
a	Machine index in the assembly stage
na	Total number of machines in the assembly stage
A	Set of machines and $A = \{1, \dots, na\}$ in the assembly stage
v	AGV index in the transportation stage
nA	Total number of AGVs in the transportation stage
V	Set of AGVs and $V = \{1, 2, \dots, nA\}$ in the transportation stage
t	AGV shipment index in the transportation stage
T	Set of the AGV shipments and $T = \{1, \dots, n\}$ in the transportation stage
Cap	The capacity of an AGV at each shipment
$O_{i,j}$	The j -th operation of job i in the processing stage
$t_{i,j,k}$	The time required for operation $O_{i,j}$ of machine k in the processing stage
ta_p	The time required to assemble the product p of the assembly machine in the assembly stage
Tr	Transportation time between processing shop and assembly shop
M	A very large positive integer
S_k	Start-up time for processing machine k
F_k	Shutdown time for processing machine k
SA_a	Start-up time for assembly machine a
FA_a	Shutdown time for assembly machine a
PE	Total processing energy consumption in the processing stage
PIE	Total idle energy consumption for processing machines in processing stage
$PE_{i,j,k}$	Processing energy consumption of operation $O_{i,j}$ on machine k

the assembly shop. In unloaded mode, the AGV returns from the assembly shop to the processing shop.

TABLE 1. (Continued.) Notations definition.

$Pidle_k$	Idle energy consumption for processing machine k
TE	Total transportation energy consumption in the transportation stage
LE	Loaded energy consumption of the AGV for traveling jobs from the machine shop to the assembly shop (loaded mode)
ULE	Unloaded energy consumption of the AGV for returning from the assembly shop to the processing shop (unloaded mode)
AE	Total assembly energy consumption in assembly stage
AIE	Total idle energy consumption for assembly machines in the assembly stage
AAE	Assembly energy consumption for each assembly machine
AWE	Idle energy consumption for each assembly machine
$X_{i,j,k}$	0-1 decision variable in the processing stage, if operation $O_{i,j}$ is selected to be processed on machine k , $X_{i,j,k} = 1$; otherwise, $X_{i,j,k} = 0$.
$Y_{i,j,i',j'}$	0-1 decision variable in the processing stage, if operation $O_{i,j}$ is processed before operation $O_{i',j'}$ on a particular machine, $Y_{i,j,i',j'} = 1$; otherwise, $Y_{i,j,i',j'} = 0$.
$XT_{i,v,t}$	0-1 decision variable in transportation stage, if job i is transported by AGV v for the t -th time, $XT_{i,v,t} = 1$; otherwise, $XT_{i,v,t} = 0$.
$YT_{v,t}$	0-1 decision variable in transportation stage, If the t -th transportation is performed by AGV v , $YT_{v,t} = 1$; otherwise, $YT_{v,t} = 0$.
$XA_{p,a}$	0-1 decision variable in the assembly stage, if product p is selected to be assembled on assembly machine a , $XA_{p,a} = 1$; otherwise, $XA_{p,a} = 0$.
$YA_{p,p'}$	0-1 decision variable in the assembly stage, if product p is transported before product p' , $YA_{p,p'} = 1$; otherwise $YA_{p,p'} = 0$.
$S_{i,j}$	continuous decision variable in the processing stage, representing the starting time of operation $O_{i,j}$.
$ST_{v,t}$	continuous decision variable in the transportation stage, representing the starting time of AGV v 's t th transport.
SA_p	continuous decision variable in the assembly stage, representing the starting time of product p .
C_{max}	continuous decision variable in the assembly stage, representing the makespan.

In assembly stage, there are na identical assembly machines. For each product, it can be assembled only when all its jobs have been transported to assembly shop. Moreover, each product can select to be assembled on only one assembly machine. The assembly machine has two modes, namely assembly mode and idle mode. In assembly mode,

the machine assembles a product. In idle mode, the machine waits for product after processing the current one.

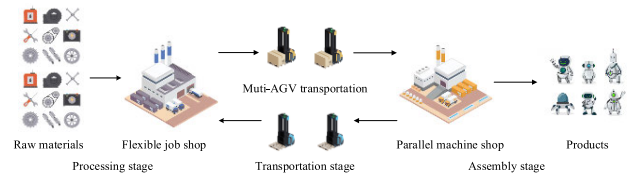


FIGURE 1. The schematic diagram of FJSP-T-A.

Moreover, FJSP-A-T is based on following assumptions:

1) Processing stage

- All jobs are independent and all machines are available at moment 0.
- A processing machine can only process one operation at a time.
- There are no processing order constraints between the operations of different jobs.
- Each operation can only be processed once.
- Transportation time of jobs between different processing machines is neglected.
- Transportation by AGVs is only possible after all operations have been completed for each job.

2) Transportation stage

- All AGVs available at moment 0.
- All AGVs are the same.
- The maximum number of jobs per transfer must not exceed the capacity of the AGV.
- The speed of each AGV is constant and the path interference of the AGVs is not considered.
- Time for loading and unloading jobs is neglected.

3) Assembly stage

- All assembly machines available at moment 0.
- There are no processing constraints between products.
- Each product can only be assembled once.
- Each assembly machine can only handle one product at a time.

A simple example is given, it includes 10 jobs, 3 processing machines, 2 AGVs and 2 assembly machines. The processing time for each operation of each job on different processing machines is shown in Table 2. In Table 2, M_i denotes the number of the processing machine. The loaded and unloaded transportation time for each AGV is 4. The load capacity of each AGV is 2. The jobs of each product and assembly times are shown in Table 3. Figure 2 illustrates a Gantt chart for one scheduling scenario for this example.

C. ENERGY CONSUMPTION OF THE FJSP-T-A SHOP

The total energy consumption consists of five parts consumed in three stages: processing energy (PE) and idle energy (PIE) in the processing stage, transportation energy (TE) in the transportation stage, and assembly energy (AE) and idle energy (AIE) in the assembly stage.

TABLE 2. The processing time of the processing machine for each operation.

Job	Operation	Processing time		
		M_1	M_2	M_3
J_1	O_{11}	3	5	4
	O_{12}	6	-	4
J_2	O_{21}	-	3	5
	O_{22}	-	4	-
J_3	O_{31}	-	4	6
	O_{32}	6	3	3
J_4	O_{41}	3	4	6
	O_{42}	5	2	-
J_5	O_{51}	2	-	4
	O_{52}	5	4	2
J_6	O_{61}	-	6	5
	O_{62}	6	5	2
J_7	O_{71}	4	5	6
	O_{72}	2	-	6
J_8	O_{81}	5	4	6
	O_{82}	2	4	2
J_9	O_{91}	4	-	5
	O_{92}	6	-	2
J_{10}	O_{101}	5	5	4
	O_{102}	5	-	5
	O_{103}	6	4	6

TABLE 3. The jobs of each product and assembly time.

Product	Job	Assembly time
P_1	J_1	6
	J_2	
	J_3	
P_2	J_4	4
	J_5	
	J_6	
P_3	J_7	8
	J_8	
	J_9	
	J_{10}	

1) PROCESSING ENERGY CONSUMPTION IN PROCESSING SHOP

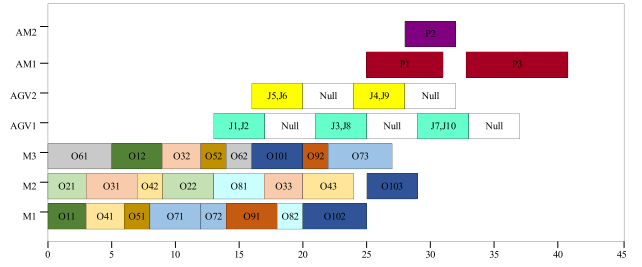
The PE indicates the energy consumed by the machines in processing mode, and it can be calculated as follows:

$$PE = \sum_{i \in I} \sum_{j \in J_i} \sum_{k \in K_{i,j}} t_{i,j,k} PE_{i,j,k} X_{i,j,k} \quad (1)$$

2) IDLE ENERGY CONSUMPTION IN PROCESSING SHOP

The PIE represents the energy consumed by machines for in idle mode, and it can be calculated as follows:

$$PIE = \sum_{i \in I} \sum_{j \in J_i} \sum_{k \in K_{i,j}} (F_k - S_k - t_{i,j,k} X_{i,j,k}) \text{Pidle}_k \quad (2)$$

**FIGURE 2.** Example of a Gantt chart.

3) TRANSPORTATION ENERGY CONSUMPTION

The TE represents the energy consumed by the AGVs for transporting jobs, and it includes two parts, namely LE and ULE. LE is the energy consumed when AGVs are in loaded mode, while ULE is the energy consumed when AGVs are in unloaded mode. TE can be calculated as follows:

$$TE = \sum_{v \in V} \sum_{t \in T} (LE + ULE) T_r Y_{v,t} \quad (3)$$

4) ASSEMBLY ENERGY CONSUMPTION IN ASSEMBLY SHOP

The AE represents the energy consumed by the assembly machine in assembly mode, and it can be calculated as follows:

$$AE = \sum_{p \in P} AAE * ta_p \quad (4)$$

5) IDLE ENERGY CONSUMPTION IN ASSEMBLY WORKSHOP

The AIE represents the energy consumed by the assembly machine in idle mode, and it can be calculated as follows:

$$AIE = \sum_{a \in A} \sum_{p \in P} (FA_a - SA_a - ta_p X_{a,p,a}) AWE \quad (5)$$

6) TOTAL ENERGY CONSUMPTION

The total energy consumption is the sum of the energy produced in each stage, and can be calculated as follows:

$$\begin{aligned} TEC &= PE + PIE + TE + AE + AIE \\ &= \sum_{i \in I} \sum_{j \in J_i} \sum_{k \in K_{i,j}} (t_{i,j,k} PE_{i,j,k} X_{i,j,k} \\ &\quad + (F_k - S_k - t_{i,j,k} X_{i,j,k}) \text{Pidle}_k) \\ &\quad + \sum_{v \in V} \sum_{t \in T} (LE + ULE) T_r Y_{v,t} \\ &\quad + \sum_{p \in P} AAE * ta_p \\ &\quad + \sum_{a \in A} \sum_{p \in P} (FA_a - SA_a - ta_p X_{a,p,a}) AWE \end{aligned} \quad (6)$$

D. OBJECTIVE FUNCTIONS AND CONSTRAINT SETS

The objective functions are as follows:

$$\begin{aligned} \min f_1 &= C_{max} \\ \min f_2 &= TEC \end{aligned} \quad (7)$$

The constraint set as follows:

$$\sum_{k \in K_{i,j}} X_{i,j,k} = 1, \forall i \in I, j \in J_i \quad (8)$$

$$S_{i,j} + pt_{i,j,k} \leq S_{i',j'} + M(3 - Y_{i,j,i',j'} - X_{i,j,k} - X_{i',j',k}), \\ \forall i \in I, i' \in I, i < i', j \in J_i, j' \in J_{i'}, k \in K_{i,j} \cap K_{i',j'} \quad (9)$$

$$S_{i',j'} + pt_{i',j',k} \leq S_{i,j} + M(2 + Y_{i,j,i',j'} - X_{i,j,k} - X_{i',j',k}), \\ \forall i \in I, i' \in I, i < i', j \in J_i, j' \in J_{i'}, k \in K_{i,j} \cap K_{i',j'} \quad (10)$$

$$S_{i,j+1} \geq S_{i,j} + \sum_{k \in K_{i,j}} pt_{i,j,k} X_{i,j,k}, \forall i \in I, \\ j \in \{1, 2, \dots, n_i - 1\} \quad (11)$$

$$S_{i,j} \geq 0, \forall i \in I, j \in J_i \quad (12)$$

$$ST_{v,t} + M(1 - XT_{i,v,t}) \geq S_{i,n_i} + \sum_{k \in K_{i,n_i}} pt_{i,n_i,k} X_{i,n_i,k}, \\ \forall j \in J, v \in V, t \in T \quad (13)$$

$$\sum_{v \in V} \sum_{t \in T} XT_{i,v,t} = 1, \forall i \in I \quad (14)$$

$$\sum_{i \in I} XT_{i,v,t} \geq \sum_{i \in I} XT_{i,v,t+1}, \forall v \in V, t \in \{1, \dots, n-1\} \quad (15)$$

$$\sum_{i \in I} XT_{i,v,t} \leq Cap, \forall v \in V, t \in T \quad (16)$$

$$ST_{v,t+1} \geq ST_{v,t} + 2Tr, \forall v \in V, t \in \{1, \dots, n-1\} \quad (17)$$

$$ST_{v,t} \geq 0, \forall v \in V, t \in T \quad (18)$$

$$SA_p + M(1 - XT_{i,v,t}) \geq ST_{v,t} + Tr, \forall v \in V, t \in T, \\ p \in P, i \in PI_p \quad (19)$$

$$\sum_{a \in A} XA_{p,a} = 1, \forall p \in P \quad (20)$$

$$SA_p + ta_p \leq SA_{p'} + M(3 - YA_{p,p'} - XA_{p,a} - XA_{p',a}), \\ \forall p \in P, p' \in P, p < p', a \in A \quad (21)$$

$$SA_{p'} + ta_{p'} \leq SA_p + M(2 + YA_{p,p'} - XA_{p,a} - XA_{p',a}), \\ \forall p \in P, p' \in P, p < p', a \in A \quad (22)$$

$$SA_p + ta_p \leq C_{max} \quad (23)$$

$$F_k + M(1 - X_{i,j,k}) \geq S_{i,j} + t_{i,j,k}, \forall i \in I, j \in J_i, k \in K_{i,j} \quad (24)$$

$$S_k \leq S_{i,j} + M(1 - X_{i,j,k}), \forall i \in I, j \in J_i, k \in K_{i,j} \quad (25)$$

$$S_k \leq F_k, \forall k \in K \quad (26)$$

$$S_k \geq 0, \forall k \in K \quad (27)$$

$$FA_a + M(1 - XA_{p,a}) \geq S_p + ta_p, \forall a \in A, p \in P \quad (28)$$

$$SA_a \leq S_p + M((1 - XA_{p,a})), \forall a \in A, p \in P \quad (29)$$

$$SA_a \leq FA_a, \forall a \in A \quad (30)$$

$$SA_a \geq 0, \forall a \in A \quad (31)$$

$$YT_{v,t} \geq XT_{i,v,t}, \forall i \in I, v \in V, t \in T \quad (32)$$

$$X_{i,j,k,v} \in \{0, 1\}, \forall i \in I, j \in J_i, k \in K_{i,j}, v \in V \quad (33)$$

$$Y_{i,j,i',j'} \in \{0, 1\}, \forall i \in I, i' \in I, i < i', j \in J_i, j' \in J_{i'} \quad (34)$$

$$XT_{i,v,t} \in \{0, 1\}, i \in I, v \in V, t \in T \quad (35)$$

$$YT_{v,t} \in \{0, 1\}, v \in V, t \in T \quad (36)$$

$$XA_{p,a} \in \{0, 1\}, \forall p \in P, a \in A \quad (37)$$

$$YA_{p,p'} \in \{0, 1\}, \forall p \in P, p' \in P, p < p' \quad (38)$$

where, constraint set (8) indicates that in the processing stage, the operation can only be selected to be processed on one processing machine. The constraint sets (9) and (10) denote the sequential processing order constraints for different operations scheduled on the same processing machine in the processing stage. In particular, when $O_{i,j}$ and $O_{i',j'}$ are selected to be processed on processing machine k , i.e., $X_{i,j,k} = 1$ and $X_{i',j',k} = 1$, respectively. If $Y_{i,j,i',j'} = 1$, constraint set (9) constrains $O_{i',j'}$ to be processed after $O_{i,j}$ and constraint (10) is relaxed; if $Y_{i,j,i',j'} = 0$, constraint set (10) constrains $O_{i',j'}$ to be processed before $O_{i,j}$ and constraint set (9) is relaxed. When $O_{i,j}$ and $O_{i',j'}$ are not processed on the same processing machine, the constraint sets (9) and (10) are relaxed. Constraint set (11) indicates that the $O_{i,j+1}$ can be started in the processing stage only when the $O_{i,j}$ has been processed. Constraint set (12) constrains the range of values of the decision variable $S_{i,j}$, i.e., all jobs are processable at moment 0. Constraint set (13) is a bridge between the processing stage and the transportation stage, and is used to constrain jobs to be transported only if the processing is completed in the processing stage. In other words, if job i is transported by AGV $_v$ at the t th time, i.e., $XT_{i,v,t} = 1$, then constraint set (13) constrains $ST_{v,t}$ to be no less than the completion time $S_{i,n_i} + \sum_{k \in K_{i,n_i}} pt_{i,n_i,k} X_{i,n_i,k}$ of the last process of job i . Constraint set (14) indicates that any job can only be transported by one AGV once in the transportation stage. Constraint set (15) indicates that in the transportation stage, the AGV must transport the jobs sequentially, i.e., there must not be an unloaded travel situation from the processing stage to the assembly stage in the middle of the two transportation processes. Constraint set (16) indicates that the maximum number of jobs to be transported by the AGV at a time cannot exceed the capacity of the AGV in the transportation stage. Constraint set (17) represents the relationship between two transportation start times consecutively by the AGV in the transportation stage, i.e., the start time $ST_{v,t+1}$ of the next time is not less than the sum of the current start time $ST_{v,t}$ and the round-trip time $2Tr$. Constraint set (18) denotes the decision variable $ST_{v,t}$ takes a range of values, i.e., AGVs are available at moment 0. Constraint set (19) connects the transportation stage to the assembly stage. It ensures that assembly of a product cannot start until all the jobs for that product have been delivered to the assembly shop. Constraint set (20) indicates that a product can be selected to be assembled on only one assembly machine in the assembly stage. The constraint sets (21) and (22) indicate the sequential relationship between different products assigned to the same assembly machine in the assembly stage. When both products p and p' are chosen to be assembled on assembly machine a , i.e., $XA_{p,a} = 1$ and

$XA_{p',a} = 1$. If $YA_{p,p'} = 1$, constraint set (21) constrains product p to be assembled before product p' , and the start assembly time $SA_{p'}$ of product p' is not less than the assembly completion time $SA_p + ta_p$ of the product p , and constraint set (22) is relaxed and constant. If $YA_{p,p'} = 0$, constraint set (22) constrains product p to be assembled after product p' , and the start assembly time SA_p of product p is not less than the assembly completion time $SA_{p'} + ta_{p'}$ of product p' , and constraint set (21) is relaxed and constant. If products p and p' are not assembled on the same assembly machine, the constraint sets (21) and (22) are both relaxed and constant. Constraint set (23) represents the makespan constraint, i.e., the makespan is not less than the assembly completion time of all products. Constraint set (24) indicates that in the processing stage, a processing machine must wait until all operations performed on this processing machine have been fully processed before it can be shut down. Constraint set (25) indicates that in the processing stage, a processing machine must wait until an operation is ready to be processed before it is turned on. Constraint set (26) constrains the range of values of F_k , i.e., the processing machine is turned off only when it is turned on. Constraint set (27) constrains the range of values of S_k , i.e., all processing machines are available at moment 0 for. Constraint set (28) indicates that in the assembly stage, an assembly machine can only be shut down after all products on this assembly machine have been fully assembled. Constraint set (29) indicates that in the assembly stage, an assembly machine is not started until a product is available for assembly. Constraint set (30) constrains the range of values of FA_a , i.e., the assembly machine can only be turned on before it can be turned off. Constraint set (31) constrains the range of values of SA_a , i.e., all assembly machines available at moment 0. Constraint set (32) denotes the relationship of decision variables $YT_{v,t}$ and $XT_{i,v,t}$. Constraint sets (33)-(38) defines the 0-1 decision variables.

IV. ALGORITHM

A. CO-EVOLUTIONARY ALGORITHM

Co-evolutionary algorithm (CEA) is an evolutionary computational algorithm based on group intelligence for solving complex optimization problems [40]. The core idea of the algorithm is to decompose the problem into multiple sub-problems, then optimize these sub-problems separately, and gradually improve the quality of the overall solution by collaborative work. In CEA, each subproblem is solved by a separate evolutionary computational module, which typically use GA, PSO, or other optimization techniques. The optimization result of each subproblem affects the quality of the overall solution. Thus, collaboration between different subproblems is needed to co-evolve in order to find the global optimal solution [41].

B. PROPOSED ALGORITHM MPCEA

1) GENERAL FRAMEWORK OF MPCEA

The framework of MPCEA is shown in Figure 3. The steps of MPCEA are given below:

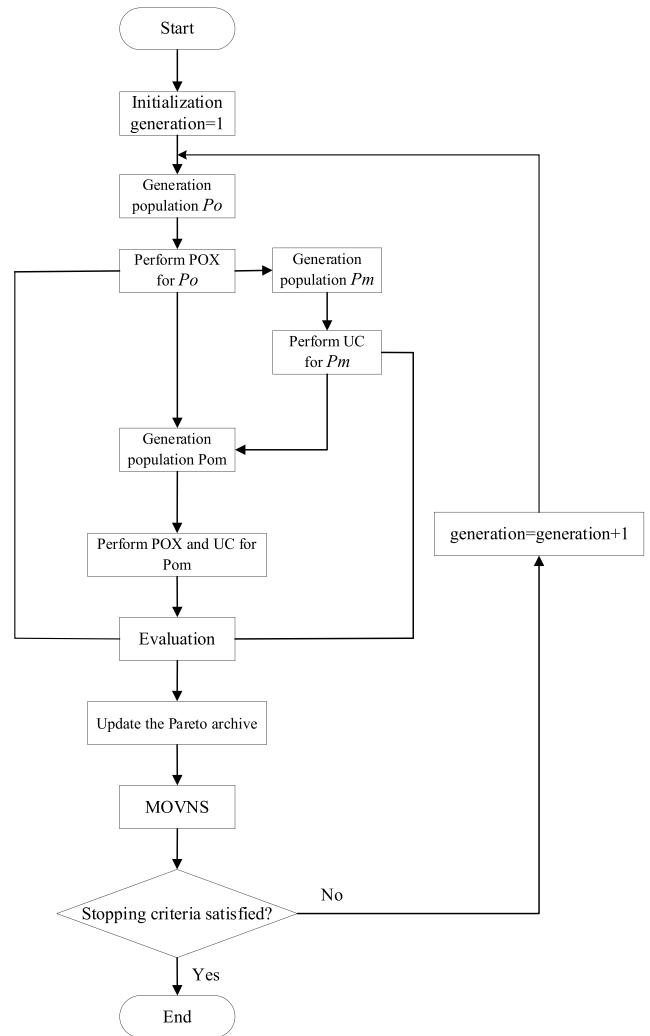


FIGURE 3. The general framework of the proposed MPCEA.

Step 1 Initialization: Generate the initial population P . Set the population size N and the number of MOVNS T_{vms} . Additionally, the population is initialized. Implement the non-dominated sorting algorithm on the population and then update the external Pareto archive set with the Pareto solutions in the first Pareto level front. Steps 2–10 are repeated until the stopping criterion is satisfied. Set $generation=1$.

Step 2 Generation population P_O : Copy the individuals from the initial population P to the population P_O .

Step 3 Perform POX for P_O : Perform POX for P_O according to the method described in Section IV-B6. Then denote P_O as P'_O .

Step 4 Generation population P_M : Copy the individuals from population P'_O to the population P_M .

Step 5 Perform UC for P_M : Perform UC for P_M according to the method described in Section IV-B6. Then denote P_M as P'_M .

Step 6 Generation population P_{OM} : Select individuals from P'_O and P'_M according to the method described in Section IV-B8 and add them to P_{OM} .

Step 7 Perform POX and UC for P_{OM} : Perform POX and UC for P_{OM} according to the method described in Section IV-B6. Then denote P_{OM} as P'_{OM} .

Step 8 Evaluation: Individuals in P'_O , P'_M , and P'_{OM} were evaluated.

Step 9 Update the Pareto archive: Update the Pareto archive set according to the method described in 4.2.7.

Step 10 MOVNS: A Local search for Pareto solutions in the Pareto archive set is performed according to the method described in 4.2.9.

2) ENCODING

To solve the FJSP-T-A, we make two main decisions in terms of encoding, corresponding to two subproblems: the order of processing the operations and the selection of processing machines. These decisions are represented by two encoded strings, as shown in Figure 4. In this case, the processing sequence code relates to all operations of all jobs, e.g., (1, 2, 1, 3, ...) indicates that the operations ($O_{1,1}$, $O_{2,1}$, $O_{1,2}$, $O_{3,1}$, ...) are executed sequentially in order, and the 2nd 1 in the encode indicates the second operation of job 1. The processing machine selection is encoded according to the job serial number and operation sequence number, e.g., (1, 3, 2, 3, ...) indicates the processing machine number assigned to the 4 operations of Job 1. It is important to note that the assigned processing machine number does not represent the corresponding operation of the selected processing machine, but rather an index to the set of candidate processing machines. Both strings are equal in length and equal to the total number of operations for all jobs.

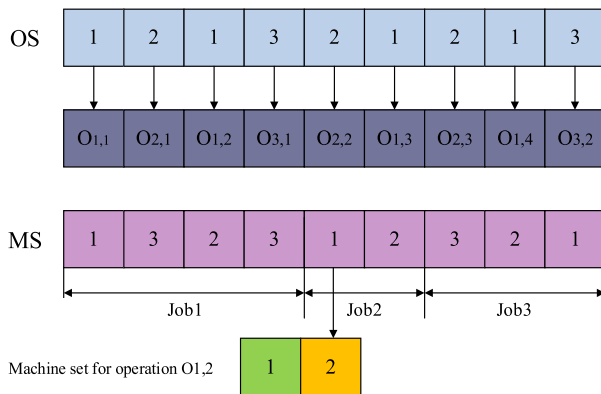


FIGURE 4. Encoding diagram.

3) DECODING

In MPCEA, we design an active decoding rule. When jobs are completed, an idle AGV is prioritized for transportation based on the order of processing completion time, from smallest to largest. If there are at least two idle AGVs, one AGV is randomly selected. If the number of untransported jobs is greater than or equal to the maximum load capacity of an AGV, the AGV must be fully loaded for each transportation. A product can only be assembled after all jobs contained

within it have arrived at the assembly shop. Following the chronological order in which products become ready for assembly, a free assembly machine is prioritized for use. If there are at least two free assembly machines, one assembly machine is randomly selected.

4) POPULATION INITIALIZATION

In this paper, we apply co-evolutionary methods to chromosomes and populations. Based on the encoding method, we divide the population into the operations sequencing population P_O of the operations and the processing machine selection population P_M of the operations. To increase the diversity of individuals, we also add the population P_{OM} . The populations of P_O , P_M and P_{OM} are equal in size.

5) POPULATION CLASSIFICATION

In this paper, we apply co-evolutionary methods to chromosomes and populations. Based on the encoding method, we divide the population into the operations sequencing population P_O of the operations and the processing machine selection population P_M of the operations. To increase the diversity of individuals, we also add the population P_{OM} . The populations of P_O , P_M and P_{OM} are equal in size.

6) Crossover

In order to more efficiently search, we use precedence operation crossovers (POX) for P_O . Figure 5 gives an example of POX where P1 and P2 are defined as two parents and O1 and O2 are defined as two children. For P_M , uniform crossovers (UC) are used. Figure 6 gives an example of UC, where P1 and P2 are defined as two parents, O1 and O2 are defined as two children, and R is a randomly generated string of binary numbers of length equal to the total number of operations. The N_{oper} denotes the total number of operations.

The POX steps are as follows:

Step 1: Randomly divide jobs into two parts, set 1 and set 2.

Step 2: Copy the elements belonging to set 1 from parent P1/P2 to child O1/O2 according to the original position.

Step 3: Copy the remaining uncopied elements in the parent P1/P2 into O2/O1 in order.

The UC steps are as follows:

Step 1: Generate N_{oper} random binary numbers.

Step 2: Iterate through the string, O1 inherits the processing machine number of P1 in the position where the element is 1 and the processing machine number of P2 in the position where the element is 0.

Step 3: Generate O2 in the same way as Step2.

7) THE PARETO ARCHIVE SET

A pareto archive set is used to store non-dominated solutions. As the algorithm iterates, the Pareto archive set is being updated. At each iteration, three sub-populations are evaluated and the generated solutions are involved in updating the Pareto archive set. If the solution is a non-dominated solution, it is added to the Pareto archive set. If an existing solution in the Pareto archive set is dominated, it is removed.

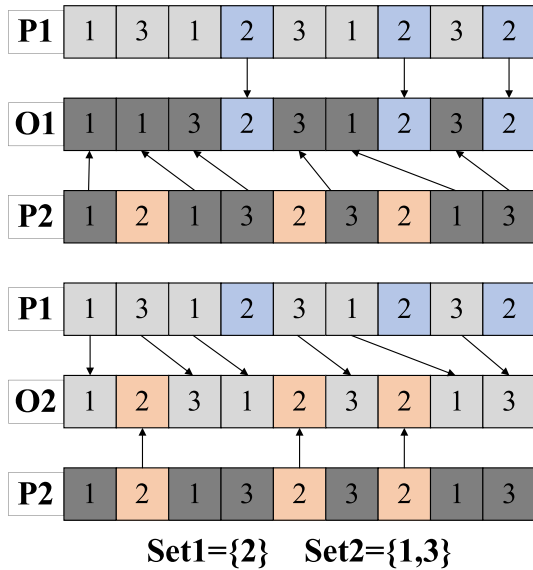


FIGURE 5. An example of POX.

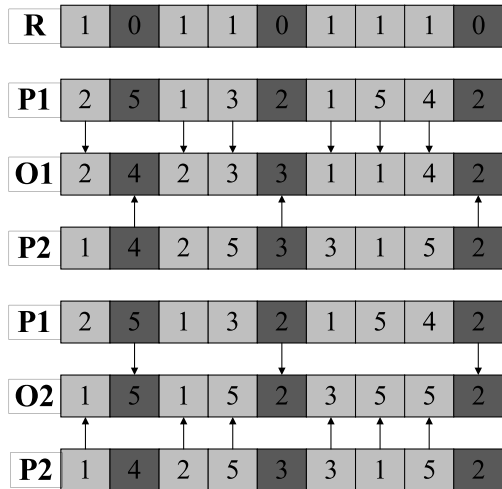


FIGURE 6. An example of UC.

8) THE SELECTION STRATEGY FOR INDIVIDUALS IN P_{OM}

For a single objective problem, elite individuals in the population can be selected by evaluating the objective value. However, for multi-objective problems, there is often a mutually exclusive relationship between the objectives, and they have different scales, which can lead to an unreasonable assessment of fitness values. Therefore, a direct judgment based on fitness values cannot effectively select the best individuals in the population. For this reason, we used the Max-min method for normalization. The calculated formula is as follows,

$$f'_i = \frac{f_i - f_{min}}{f_{max} - f_{min}} \quad (39)$$

where, f_i is the original target value, f'_i is the objective value after normalization of f_i , f_{max} and f_{min} are the maximum and minimum values in the original objective value, respectively.

Algorithm 1 Algorithm for Selecting Strategy

Input: a set of individuals $population_i$ for $i = 1, 2, \dots, N$ in P_O .

a set of individuals $population_j$ for $j = 1, 2, \dots, N$ in P_M .

Output: a fine population P_{OM}

1: The individual objective values ($Cmax_i$ and TEC_i) in P_O and P_M are normalized.

2: Calculating α_i values and β_j values.

3: Sort by α_i values in ascending order of P_O 's individuals.

Sort by β_j values in ascending order of P_M 's individuals.

4: The top 50% individuals of each P_O and P_M are selected to form P_{OM} .

With this approach, each objective has a value domain range period of [0, 1]. Two different objective values are placed on the same scale and can be compared directly.

In order to select relatively good individuals, the α value of individuals in P_O and the β value of individuals in P_M are calculated when the number of iterations is odd. The individuals in P_O and P_M are then sorted in ascending order according to the values of α and β , respectively. The top 50% of individuals from each population were selected to form the population P_{OM} . When the number of iterations is even, the β value of individuals in P_O and the α value of individuals in P_M are calculated, and the subsequent operations are the same as when iterations is odd. The α value and β value are calculated as follows,

$$\alpha_i = \frac{Cmax'_i}{TEC'_i} \quad (40)$$

$$\beta_j = \frac{TEC'_j}{Cmax'_j} \quad (41)$$

where, $Cmax'_i$ and TEC'_i are $Cmax_i$ and $Cmax_i$ normalized values, respectively. A large value of α indicates that makespan is the dominant factor. A large value of β indicates that TEC is the dominant factor. The steps for selecting the strategy are detailed in Algorithm 1 (when the number of iterations is odd).

9) MULTI-OBJECTIVE VARIABLE NEIGHBORHOOD SEARCH (MOVNS)

In order to improve the development capability of MPCEA, a MOVNS algorithm is designed and embedded into MPCEA. The steps of the MOVNS algorithm are detailed in Algorithm 2. In MOVNS, six neighboring structures are used to locally search all the nondominated solutions in the Pareto archive set. The six neighborhood structures are described below:

N1: In OS strings, two different positions are randomly selected and the genes at these two positions are exchanged. As shown in Figure 7(a).

N2: In OS strings, two different positions are randomly selected, and the genes between the two positions are shifted back one by one in order. As shown in Figure 7(b).

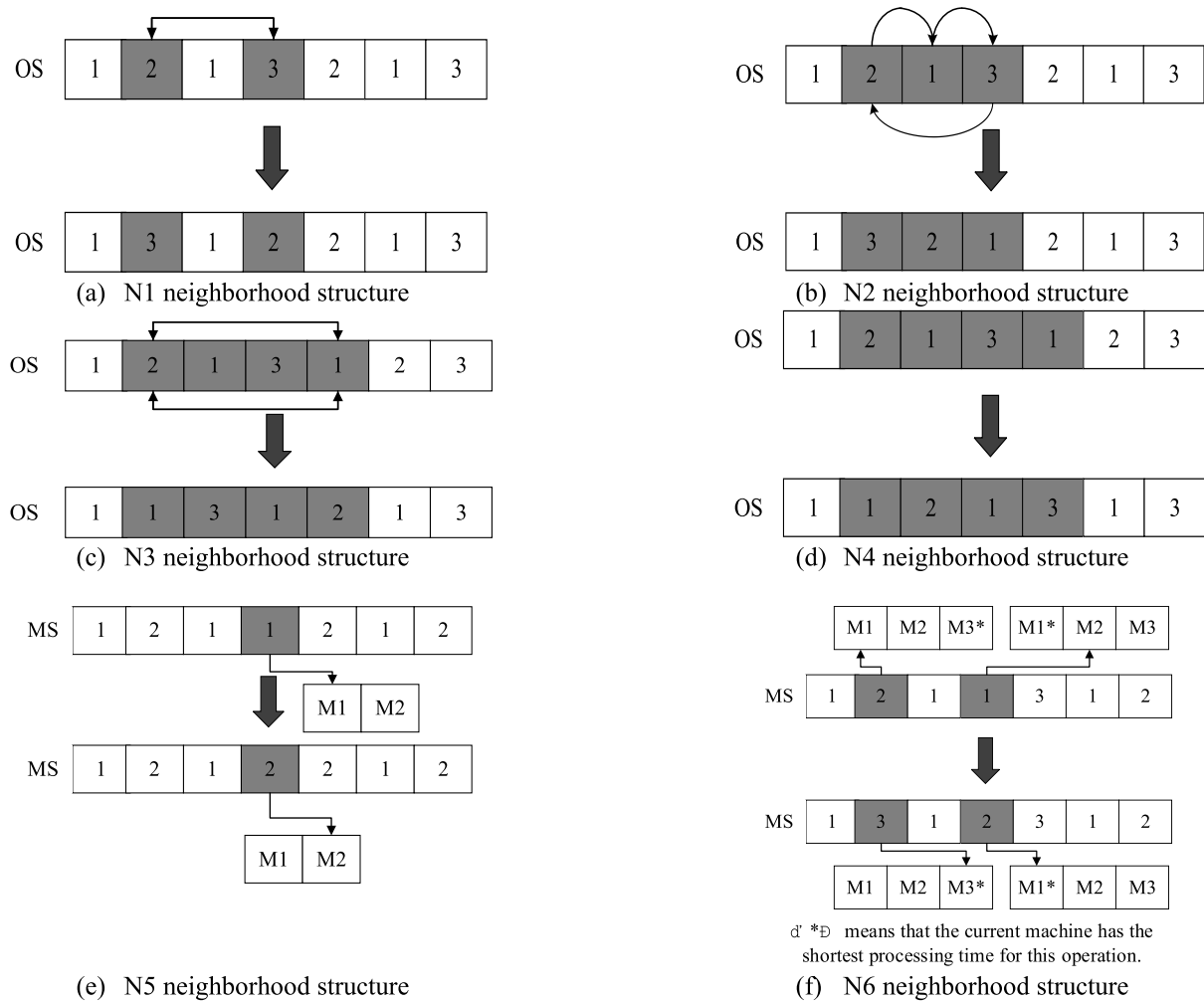


FIGURE 7. Six kinds of neighborhood structures of MOVNS algorithm.

N3: In OS strings, two different positions are randomly selected and the genes between the two positions are sorted in reverse order. As shown in Figure 7(c).

N4: In OS strings, two different positions are randomly selected and the genes between the two positions are randomly disrupted. As shown in Figure 7(d).

N5: In MS strings, randomly select a location to change its processing machine assignment. As shown in Figure 7(e).

N6: In MS strings, two different positions are randomly selected, and the processing machine at that position is replaced with the one that has the shortest current processing time. As shown in Figure 7(f).

V. COMPARISON OF EXPERIMENTS AND ANALYSIS OF RESULTS

The proposed algorithms and all the comparison algorithms are implemented on a computer with a 12th generation Intel (R) Core (TM) i7-12700 CPU @ 2.10 GHz and 16 GB of RAM, and compiled on Visual Studio 2019 using Visual Studio 2019 in C++. The proposed MILP model is

executed in IBM CPLEX Studio IDE 22.1.1 with using OPL modeling language. The solving method used is branch-and-cut [19]. The branch-and-cut method is the combination of cutting plane and branch-and-bound methods. The instances MFJSTA01-10 and MKTA01-10 are generated by extending classical MFJS01-10 [42] and MK01-10 [43]. The processing powers of the processing machines are randomly generated in the interval $[0, 8]$ and the idle powers are randomly generated in the set $\{1, 2, 3\}$. The load powers of the AGVs are 2 and the unload powers of the AGVs are 1. The assembly powers of the assembly machines are 5 and the idle powers of the assembly machines are 2. The maximum load of each AGV is 2. The number of the assembly machines is set to 2. For the comparison of the meta-heuristic algorithms and the validation of the strategies, each algorithm is repeated ten times, and the set of non-dominated solutions of these repeated runs are collected and compared. The stopping time for each algorithm is $2N_{oper}$ seconds, and the stopping time of MILP is set to 600 seconds.

Algorithm 2 Algorithm for MOVNS

Input: a solution x_i , $i = 1, 2, \dots, X$. X is the size of archive set
 a set of neighborhood structures N_k for $k = 1, 2, \dots, k_{max}$ for shaking.
 a set of neighborhood structures N_l for $l = 1, 2, \dots, l_{max}$ for local search.
 Output: an improved solution x_i
 Repeat
 While $i \leq X$ Do
 For $k = 1$ To k_{max} Do
 Shaking: $x_i' = N_k(x_i)$.
 For $t = 1$ To T_{vns}
 set $l = 1$;
 while $l \leq l_{max}$ Do
 $x_i'' = N_l(x_i')$
 If (x_i'') dominate x_i'
 $x_i' = x_i''$;
 $l = 1$;
 else
 $l = l + 1$;
 end if
 end while
 end for
 If (x_i') dominate x_i
 $x_i = x_i'$;
 update achieve set and X ;
 end if
 end for
 end while

TABLE 4. Main parameters and levels.

Parameters	Levels		
	1	2	3
N	300	400	500
T_{vns}	5	10	15

A. PERFORMANCE METRICS**1) GENERATIONAL DISTANCE (GD)**

The GD denotes the average shortest Euclidean distances from all the non-dominated solutions in the set X and reference set R . The smaller the value of GD is, the better the convergence is. The GD is calculated as follows,

$$GD = \frac{\sqrt{\sum_{x_i \in X} \min_{r \in R} dis(x_i, r)^2}}{|X|} \quad (42)$$

where, $dis(x_i, r)$ denotes the Euclidean distance of the non-dominated solution x_i and the reference point r in R .

2) INVERSE GENERATIONAL DISTANCE (IGD)

IGD is similar to GD except that after calculating the Euclidean distance, it is averaged instead squared. The smaller the value of IGD is, the better the diversity and

TABLE 5. Orthogonal parameter settings for MPCEA.

Number	Levels		IGD
	N	T_{vns}	
1	1	1	3.22E-05
2	1	2	2.25E-05
3	1	3	2.05E-05
4	2	1	6.29E-06
5	2	2	4.33E-06
6	2	3	1.65E-05
7	3	1	1.24E-05
8	3	2	1.18E-05
9	3	3	5.58E-06

convergence of the algorithm is. IGD is calculated as follows,

$$IGD = \frac{\sum_{x_i \in X} \min_{r \in R} dis(x_i, r)}{|X|} \quad (43)$$

where, $dis(x_i, r)$ denotes the Euclidean distance from the nondominated solution x_i to the reference point r in R .

B. PARAMETER SETTINGS

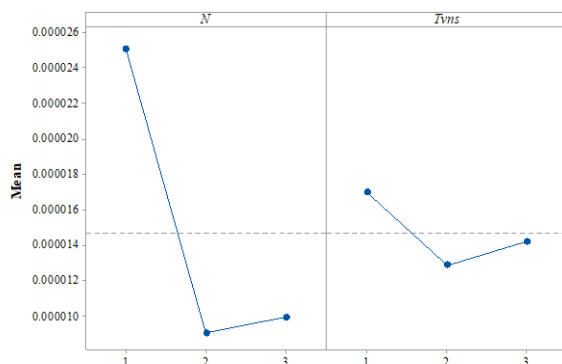
For MPCEA, there are 2 main parameters, namely the population size N and the number of MOVNS T_{vns} . In order to test the effect of parameter settings on the effectiveness of MPCEA, Taguchi experiments are conducted. The levels of the 2 parameters are shown in Table 4.

TABLE 6. Computational complexity analysis of the MILP model.

Instances	NBV	NCV	NC	CS	T(s)	Gap
MFJSTA01	206	46	477	10611.4	306.5	0
MFJSTA02	231	48	530	9340.4	600	0.02
MFJSTA03	310	53	737	11964.4	600	0.04
MFJSTA04	392	58	952	14306	600	0.13
MFJSTA05	397	58	938	14012	600	0.09
MFJSTA06	493	64	1187	16895	600	0.18
MFJSTA07	663	72	1547	22560.6	600	0.18
MFJSTA08	806	80	1760	27176.6	600	0.21
MFJSTA09	1116	92	2504	34665.6	600	0.28
MFJSTA10	1304	98	2958	65432.4	600	0.16
MKTA01	1449	97	3453	1087.4	600	0.17
MKTA02	1971	100	9894	1038.8	600	0.23
MKTA03	9254	207	27018	5953.4	600	0.3
MKTA04	2935	147	6264	2300	600	0.22
MKTA05	4260	155	10004	4496.6	600	0.25
MKTA06	9604	200	26922	2503.2	600	0.33
MKTA07	5297	161	18737	4494.8	600	0.23
MKTA08	8820	296	16343	16157.8	600	0.21
MKTA09	17298	311	42248	13986.6	600	0.25
MKTA10	21190	313	55103	13749.4	600	0.43

TABLE 7. Optimal Pareto solutions of MFJSTA01 and MFJSTA02.

MFJSTA01				MFJSTA02			
ε	C_{max}	TEC	T(s)	ε	C_{max}	TEC	T(s)
494	494	11154.2	0.22	477	477	10255.4	2.98
503	503	11139.2	0.66	493	493	9829.6	2.05
504	504	10938.8	0.69	494	494	9540	2.13
505	504	10938.8	0.7	579	579	9425.6	8.98
523	523	10842.2	0.84	580	579	9425.6	6.59
530	530	10761.4	0.8	585	585	9397.8	4.63
570	570	10670.2	1.28	633	633	9340.8	9.2
580	580	10659.8	1.06	756	756	9340.4	46.52
680	680	10611.4	3.48				
690	680	10611.4	4.5				

**FIGURE 8.** Trend of influence of each parameter of MPCEA.

Depending on the level of each parameter, experiments are conducted using an orthogonal array $L_9(3^2)$, which

consists of nine combinations of parameters at different levels. MKTA10 is tested, and the performance is evaluated with IGD. The stopping time is set to $2N_{oper}$ seconds. The comparison results are shown in Table 5.

The smaller the value of IGD, the better of the algorithm is. Figure 8 shows the trend of the levels of the factors. Therefore, the parameters are set as: $N = 400$ and $T_{vms} = 10$.

C. MILP MODEL VALIDATION

1) VALIDATION OF SINGLE-OBJECTIVE

The results of the MILP model for the single objective of minimizing the energy consumption are given in Table 6. In this table, NBV denotes the number of 0-1 decision variables, NCV denotes the number of continuous decision variables, NC denotes the number of constraints, CS denotes the best solution obtained within the constraint time, and T(s) denotes

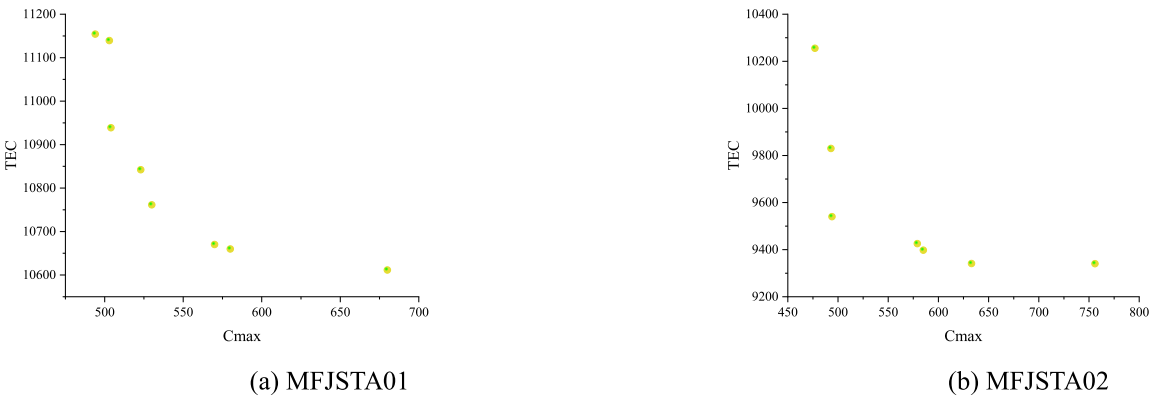


FIGURE 9. Optimal pareto front of MFJSTA01 and MFJSTA02.

TABLE 8. Comparison results.

Instances	GD		IGD	
	MPCEA	MPCEA_RS	MPCEA	MPCEA_RS
MFJSTA01	0.00E+00	6.07E-03	0.00E+00	3.69E-04
MFJSTA02	0.00E+00	3.99E-02	0.00E+00	1.12E-02
MFJSTA03	0.00E+00	0.00E+00	1.50E-04	6.73E-03
MFJSTA04	3.07E-04	3.68E-03	2.65E-06	1.44E-04
MFJSTA05	5.13E-03	7.72E-03	1.12E-04	2.98E-04
MFJSTA06	4.86E-03	2.93E-02	2.46E-03	2.67E-03
MFJSTA07	7.85E-03	5.49E-03	1.16E-03	1.97E-03
MFJSTA08	2.91E-03	3.69E-03	4.79E-04	5.81E-04
MFJSTA09	6.64E-03	3.18E-03	2.86E-04	1.23E-02
MFJSTA10	2.60E-03	1.18E-03	9.63E-04	5.99E-03
MKTA01	0.00E+00	9.50E-05	2.87E-07	1.56E-07
MKTA02	0.00E+00	2.22E-04	0.00E+00	5.32E-07
MKTA03	0.00E+00	0.00E+00	0.00E+00	0.00E+00
MKTA04	0.00E+00	2.00E-04	0.00E+00	8.64E-08
MKTA05	1.41E-04	7.53E-05	4.69E-07	1.27E-06
MKTA06	5.22E-04	9.82E-05	1.53E-07	1.36E-06
MKTA07	1.48E-05	1.61E-04	1.11E-07	4.68E-07
MKTA08	0.00E+00	9.15E-04	0.00E+00	8.37E-07
MKTA09	0.00E+00	1.74E-03	1.79E-05	2.17E-05
MKTA10	0.00E+00	8.53E-04	0.00E+00	5.10E-06



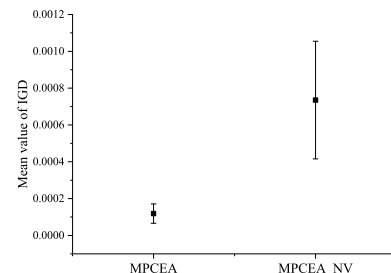
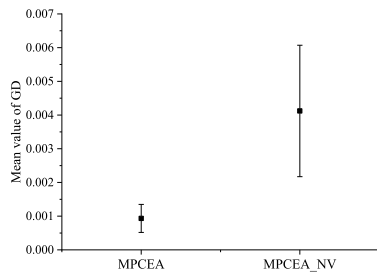
FIGURE 10. ANOVA analysis of the comparison results of MPCEA and MPCEA_RS.

the solution time. Gap denotes the tolerance between CS and the best solution (BS) that can be solved by the model, which is calculated as $Gap = (CS - BS) / BS$. If the value of Gap is 0,

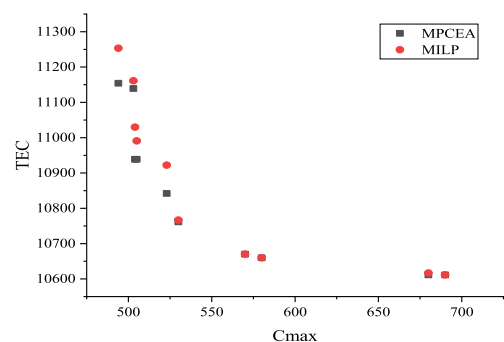
the obtained solution is optimal. In Table 6, MILP can find the optimal solution only for MFJSTA01 within 600s. This is because as the size of the problem increases, the NBV, NCV

TABLE 9. Comparison results of MPCEA and MPCEA_NV.

Instances	GD		IGD	
	MPCEA	MPCEA_NV	MPCEA	MPCEA_NV
MFJSTA01	0.00E+00	0.00E+00	0.00E+00	0.00E+00
MFJSTA02	0.00E+00	0.00E+00	0.00E+00	0.00E+00
MFJSTA03	0.00E+00	3.41E-02	2.40E-04	1.53E-03
MFJSTA04	1.17E-03	1.18E-03	1.72E-05	2.48E-04
MFJSTA05	1.75E-03	2.66E-03	1.23E-04	1.04E-04
MFJSTA06	7.35E-03	0.00E+00	4.03E-04	1.00E-03
MFJSTA07	2.63E-03	2.14E-04	9.67E-04	2.02E-03
MFJSTA08	1.84E-03	5.78E-03	6.06E-05	5.89E-03
MFJSTA09	3.75E-03	4.12E-03	2.44E-04	1.40E-03
MFJSTA10	8.77E-05	2.27E-02	3.19E-04	2.33E-03
MKTA01	4.59E-06	4.00E-05	6.59E-08	7.28E-08
MKTA02	3.19E-06	1.15E-04	6.12E-11	1.04E-07
MKTA03	0.00E+00	2.74E-05	0.00E+00	1.28E-06
MKTA04	1.28E-05	2.86E-04	1.14E-09	9.06E-07
MKTA05	6.17E-05	1.70E-04	9.70E-08	3.77E-06
MKTA06	0.00E+00	5.40E-04	0.00E+00	3.60E-06
MKTA07	0.00E+00	2.95E-04	0.00E+00	5.45E-07
MKTA08	0.00E+00	6.65E-04	0.00E+00	4.42E-07
MKTA09	0.00E+00	5.00E-03	0.00E+00	3.68E-05
MKTA10	0.00E+00	4.53E-03	0.00E+00	1.32E-04

**FIGURE 11.** ANOVA analysis of the comparison results of MPCEA and MPCEA_NV.**TABLE 10.** Pareto solutions of the MILP model and MPCEA for MFJSTA01.

Solution	MILP		MPCEA	
	C_{max}	TEC	C_{max}	TEC
1	494	11154.2	494	11253.4*
2	503	11139.2	503	11161.2*
3	504	10938.8	504	11029.8*
4	505	10938.8	505	10990.8*
5	523	10842.2	523	10922.2*
6	530	10761.4	530	10766.8*
7	570	10670.2	570	10670.2
8	580	10659.8	580	10659.8
9	680	10611.4	680	10616.8*
10	690	10611.4	690	10611.4

**FIGURE 12.** Comparison of Pareto fronts of MILP model and MPCEA for MFJSTA01.

and NC of the MILP model also increase, and the branching becomes difficult.

2) VALIDATION OF MULTI-OBJECTIVES

Table 7 lists the optimal Pareto solutions obtained using the ε -constraint method for instances MFJSTA01 and MFJSTA02. Specifically, 8 and 7 optimal Pareto solutions

are obtained for MFJSTA01 and MFJSTA02, respectively. Figure 9 shows the trend of all the Pareto solutions.

D. MPCEA VALIDATION

1) VALIDATION OF THE SELECTION STRATEGY

To verify the effectiveness of the selection strategy, we compared MPCEA with MPCEA_RS. In MPCEA_RS, P_{OM} is obtained by randomly selecting individuals from P_O and P_M .

TABLE 11. GD comparison of the MPCEA with the NSGA-II and MOEA/D.

Instances	GD		
	MPCEA	NSGA-II	MOEA/D
MFJSTA01	0.00E+00	0.00E+00	0.00E+00
MFJSTA02	0.00E+00	2.83E-02	6.11E-02
MFJSTA03	0.00E+00	0.00E+00	1.28E-02
MFJSTA04	1.37E-03	2.70E-03	2.23E-02
MFJSTA05	5.58E-03	3.01E-03	2.66E-02
MFJSTA06	4.23E-04	0.00E+00	5.40E-02
MFJSTA07	3.60E-03	1.44E-03	8.73E-02
MFJSTA08	1.64E-03	3.67E-03	1.11E-01
MFJSTA09	1.17E-03	4.97E-03	1.27E-01
MFJSTA10	7.27E-03	8.47E-03	1.48E-01
MKTA01	0.00E+00	1.45E-04	1.36E-03
MKTA02	0.00E+00	1.78E-04	1.72E-03
MKTA03	0.00E+00	1.03E-03	8.82E-03
MKTA04	0.00E+00	3.03E-04	2.83E-03
MKTA05	0.00E+00	2.78E-04	2.35E-03
MKTA06	3.59E-05	3.01E-04	5.24E-03
MKTA07	0.00E+00	7.93E-04	5.01E-03
MKTA08	0.00E+00	5.32E-03	1.96E-02
MKTA09	0.00E+00	3.75E-03	4.00E-02
MKTA10	0.00E+00	3.85E-03	2.90E-02
Mean	1.05E-03	3.42E-03	3.83E-02

TABLE 12. IGD comparison of the MPCEA with the NSGA-II and MOEA/D.

Instances	IGD		
	MPCEA	NSGA-II	MOEA/D
MFJSTA01	0.00E+00	2.27E-03	0.00E+00
MFJSTA02	0.00E+00	4.66E-03	2.79E-04
MFJSTA03	1.65E-03	4.03E-02	3.74E-03
MFJSTA04	1.29E-04	4.05E-04	1.61E-02
MFJSTA05	1.17E-03	3.18E-02	8.53E-03
MFJSTA06	2.68E-06	6.43E-04	3.89E-02
MFJSTA07	2.76E-03	1.28E-02	8.16E-02
MFJSTA08	1.38E-04	2.32E-03	1.11E-01
MFJSTA09	3.76E-05	2.65E-04	1.82E-01
MFJSTA10	2.14E-03	5.11E-04	1.76E-01
MKTA01	1.49E-07	1.60E-07	7.31E-06
MKTA02	0.00E+00	6.69E-07	9.24E-06
MKTA03	0.00E+00	2.78E-06	2.94E-04
MKTA04	4.33E-06	4.37E-07	6.02E-05
MKTA05	0.00E+00	1.42E-06	7.10E-05
MKTA06	7.47E-07	8.93E-07	2.26E-04
MKTA07	0.00E+00	4.90E-06	1.70E-04
MKTA08	0.00E+00	2.84E-05	1.76E-03
MKTA09	0.00E+00	5.22E-05	5.97E-03
MKTA10	0.00E+00	8.74E-05	5.07E-03
Mean	4.02E-04	4.81E-03	3.16E-02

The results of the GD and IGD are given in Table 8. From Table 8, it can be seen that for GD values, MPCEA is better than MPCEA_RS for 15 instances. For IGD values, MPCEA

is better than or equal to MPCEA_RS for 19 instances. Figure 10 gives a plot of the mean values of the confidence intervals of the ANOVA analysis of the indicators GD and



FIGURE 13. ANOVA analysis of the comparison results of MPCEA, NSGA-II and MOEA/D.

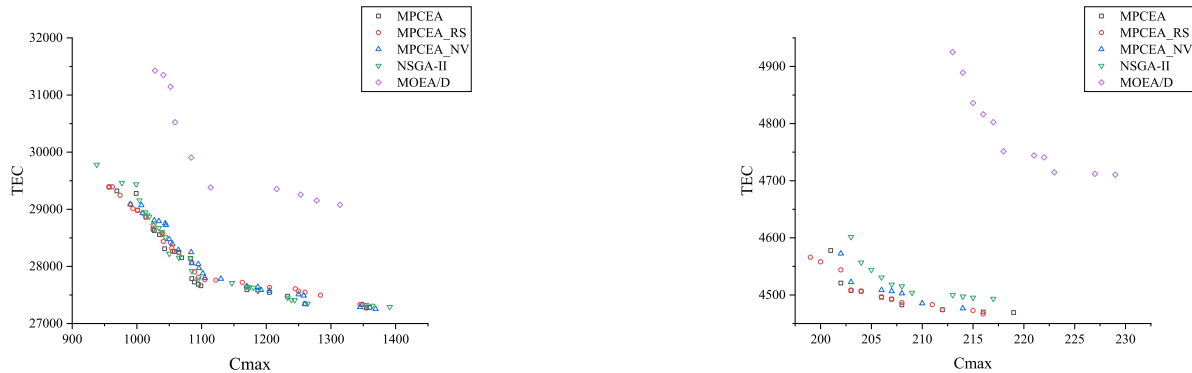


FIGURE 14. Pareto fronts of different algorithms for MFJSTA08 and MKTA05.

IGD, and it can be observed that the confidence intervals of MPCEA do not overlap with those of MPCEA_RS. This means that MPCEA outperforms MPCEA_RS statistically. In summary, it can be concluded that the selection strategy is effective.

2) VALIDATION OF THE MOVNS

To verify the validity of MOVNS, MPCEA is compared with MPCEA_NV, which is the version without MOVNS. The results for GD and IGD are given in Table 9. For GD, MPCEA is better than or equal to MPCEA_NV for all instances. For IGD, MPCEA is better than or equal to MPCEA_NV for 19 instances. Figure 11 gives a plot of the mean values of the confidence intervals of the ANOVA analysis of the indicators GD and IGD. In Figure 11, MPCEA confidence intervals do not overlap with the confidence intervals of MPCEA_NV. This means that MPCEA outperforms MPCEA_NV statistically. In summary, it can be concluded that the MOVNS is effective.

3) COMPARISONS OF THE MPCEA AND MILP MODEL

The Pareto fronts for MILP and MPCEA for MFJSTA01 are shown in Figure 12. Table 10 lists the Pareto solutions for MILP and MPCEA with the same makespan. In Table 10, the values marked with “*” are not optimal. In Table 10, MILP and MO-HSFLA can obtain the same TEC for 3 solutions. The MILP model outperforms MPCEA in terms of TEC for the solutions with makespan of 494, 503, 504, 505, 523, 530 and 680. This is because MPCEA is only an approximation algorithm, which does not guarantee to find optimal

solutions. However, for large-sized instances, MPCEA has the advantage of being able to efficiently obtain approximate optimal solutions, while the MILP model may not be able to find feasible solutions.

4) COMPARISON OF MPCEA WITH STATE-OF-ART ALGORITHMS

To validate the effectiveness of MPCEA, we compare it with two state-of-art multi-objective algorithms, namely NSGA-II and MOEA/D [19]. The population size, stopping time, encoding and decoding for NSGA-II and MOEA/D are the same as for MPCEA. For NSGA-II, the crossover operator is the same as MPCEA, and it uses the inversion variant for OS strings and the exchange variant for MS strings. The selection operator is a binary tournament selection, the crossover probability is 0.8, and the mutation probability is 0.1. For MOEA/D, the number of the sub-problems (weight vectors) is equal to the population size, and the neighborhood size of each weight vector is set as 10. Additionally, the crossover operator is the same as MPCEA, and the Tchebycheff approach is used as the aggregation methodology.

Tables 11 and 12 give the comparisons of GD and IGD respectively. In Table 11, for GD, MPCEA performs better than or equal to NSGA-II and MOEA/D for 17 instances. In Table 12, for IGD, MPCEA performs better than or equal to NSGA-II and MOEA/D for 18 instances. The proposed MPCEA can obtain the minimum mean GD and IGD. In Figure 13, MPCEA confidence intervals do not overlap with the confidence intervals of NSGA-II and MOEA/D. This means that MPCEA outperforms NSGA-II

and MOEA/D statistically. In summary, it can be concluded that the MPCEA is effective. In order to visualize the differences between the different algorithms more intuitively, the Pareto solutions for two different-scaled instances are shown in Figure 14.

VI. CONCLUSION AND FUTURE WORK

In this paper, the FJSP-T-A is studied with the objectives of minimizing the makespan and the TEC. A MILP model and an efficient MPCEA algorithm is designed. In MPCEA, encoding, decoding, initialization method, and crossover operator are designed in detail. Specifically, a selection strategy to select relatively better individuals is designed to improve the convergence speed of the algorithm. In order to improve the local search ability of the algorithm, a MOVNS algorithm is proposed. The effectiveness of the MILP model and proposed algorithms is verified through experiments. The results show that the MILP model can obtain the optimal Pareto front for small-sized instances. The selection strategy and MOVNS are effective. The results show that the mean GD and mean IGD values of MPCEA are $1.05\text{E-}03$ and $4.02\text{E-}04$ respectively, while the mean GD and mean IGD values of NSGA-II are $3.42\text{E-}03$ and $4.81\text{E-}03$ respectively, and those of MOEA/D are $3.83\text{E-}02$ and $3.16\text{E-}02$ respectively.

In future work, we will consider solving the distributed FJSP [44], and then further solving the distributed FJSP-T-A. We will consider more complex integrated scheduling problems and apply the solution ideas to similar integrated scheduling problems. We will also consider adding constraints to our FJSP-T-A, which is a challenge and an opportunity for research. Besides, we improve the method of solving the FJSP problem or merge it with other effective methods to study and solve some other scheduling problems, such as the flow shop scheduling problem, the distributed flow shop group scheduling problem, and so on.

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